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F - Max Combo

Editorial (/contests/abc415/tasks/abc415_f/editorial)

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Time Limit: 4 sec / Memory Limit: 1024 MiB

Score : 525 points

Problem Statement

There is a string S of length N consisting of lowercase English letters. Process a total of Q queries as follows:

- Type 1 : Change the i -th character of S to x .
- Type 2 : Let t be the substring of the current S from the l -th character through the r -th character. Find $f(t)$ defined as follows for this string:
 - $f(t)$ is the maximum length of consecutive identical characters in t .
 - More precisely, when choosing integers a, b such that $1 \leq a \leq b \leq |t|$ and all characters from the a -th through the b -th character of t are equal, $f(t)$ is the maximum possible value of $b - a + 1$.
 - For example, $f(\text{aaaccbbbb}) = 4, f(\text{bbaaabb}) = 3, f(\text{x}) = 1$.

Constraints

- N is an integer between 1 and 5×10^5 , inclusive.
- S is a string of length N consisting of lowercase English letters.
- Q is an integer between 1 and 5×10^5 , inclusive.
- Type 1 queries satisfy the following constraints:
 - i is an integer between 1 and N , inclusive.
 - x is a lowercase English letter.
- Type 2 queries satisfy the following constraints:
 - l, r are integers satisfying $1 \leq l \leq r \leq N$.

Input

The input is given from Standard Input in the following format:

```
N Q
S
Query1
Query2
⋮
QueryQ
```

Here, Query_i represents the i -th query.

Type 1 queries are given in the following format:

```
1 i x
```

Type 2 queries are given in the following format:

```
2 l r
```

Output

Every time a type 2 query appears, output the answer on one line.

The use of fast input and output methods is recommended because of potentially large input and output.

Sample Input 1

Copy

Copy

```
10 5
babaacczcc
2 1 4
1 3 a
2 1 10
1 8 c
2 1 10
```

Sample Output 1

Copy

Copy

```
1
4
5
```

This input contains five queries.

- Initially, the string S is babaacczcc.
- The 1st query is type 2 with $l = 1, r = 4$.
 - The extracted string is baba, and $f(\text{baba}) = 1$.
- The 2nd query is type 1 with $i = 3, x = \text{a}$.
 - The string S after the change is baaaacczcc.
- The 3rd query is type 2 with $l = 1, r = 10$.
 - The extracted string is baaaacczcc, and $f(\text{baaaacczcc}) = 4$.
- The 4th query is type 1 with $i = 8, x = \text{c}$.
 - The string S after the change is baaaaccccc.
- The 5th query is type 2 with $l = 1, r = 10$.
 - The extracted string is baaaaccccc, and $f(\text{baaaaccccc}) = 5$.

Sample Input 2

Copy

Copy

```
1 1
a
1 1 z
```

Sample Output 2

Copy

Copy

The output may be empty.

Language

C++ 20 (gcc 12.2)

Source Code

1 |

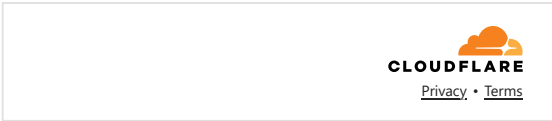
Open File

Customize

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Auto Height

* at most 512 KiB



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